EXERCISE 3 · ANOVA USING MIXED-EFFECTS MODELS

Exercise 3 in R on your Mac

The lazy recipe: put two files in a folder, install four packages, run the script, turn it into a PDF, upload. Every answer is already in the script.

CHECK **The short version**

- You hand in **one PDF** on Moodle, before **05.11.2026, 23:59**. You make it from the R script exactly like in Exercise 2 (Compile Report, then Safari, then Save as PDF).
- This time there is **no template script**, so you get the finished one: `Ex3_Alexander_Konvalina.R`. It has all the code from the exercise sheet in order, plus all 6 answers as `#` comments.
- **Don't copy code out of this PDF.** Copying from a PDF on a Mac can break underscores (like what happened with Q6 last time). Just use the `.R` file.
- Tested with R 4.6.1 and the newest packages (lme4 2.1.0), in a copy of your folder `Documents/ETH/HS26/Experimental design/Ex3`. It runs from top to bottom without errors.
- About 30 minutes in total.

Q	Pts	What it asks	The answer in one line
Q1	2	residual plots: normal? homoscedastic?	roughly yes, no strong deviation, only 2 outliers (B97 under low N)
Q2	2	why are A310, NLevelH, Replicate1 missing? + 2 yields	first level of each factor is set to 0. A310 low N = 103.11, EC169 high N = 147.76 dt/ha
Q3	1	mean of the 6 main-plot effects	0, because random effects have mean 0
Q4	1	the 4 variances	genotype 54.5, G×N 0, main plot 208.1, residual 167.5
Q5	2	σ / σ² table, which are significant, same as fixed?	genotype and main plot significant, G×N not. yes, same as with fixed genotypes
Q6	2	BLUE vs BLUP regression	R^2 = 1, slope 0.661 (= heritability), ranges 33.45 vs 22.12

1. Set up the folder (once)

1. In Finder, go to *Documents → ETH → HS26 → Experimental design*. Right-click → **New Folder** and name it exactly `Ex3`.
2. Put **two files** into `Ex3`: `DelleyDef.txt` (the data, from Moodle) and `Ex3_Alexander_Konvalina.R` (the finished script I gave you).
3. Check the data file name. It must be exactly `DelleyDef.txt`. If your Mac called it `DelleyDef (1).txt` or similar, rename it. Don't open and save it with Excel or Numbers, because that can change the file.

```
Documents/ETH/HS26/Experimental design/
└── Ex3/
    ├── DelleyDef.txt
    └── Ex3_Alexander_Konvalina.R
```

TIP **Why this exact folder?**

The first lines of the script are the `setwd(...)` line from the exercise sheet, already filled in with this folder: `setwd("~/Documents/ETH/HS26/Experimental design/Ex3")`. If you put `Ex3` somewhere else, change that one line to your path.

2. Install the 4 packages (once)

Open RStudio. Click into the **Console** (bottom left, where the `>` is) and type these 4 lines one by one, each followed by **Enter**. They are the lines from section 6.2 of the exercise sheet.

```r
install.packages("lme4")
install.packages("lmerTest")
install.packages("emmeans")
install.packages("pbkrtest")
```

If RStudio asks...	answer
"Do you want to install from sources the packages which need compilation? (Yes/no/cancel)"	type `no`, Enter
"Would you like to use a personal library instead?"	type `yes`, Enter
"Updating loaded packages. Restart R prior to install?"	click **Yes**

In the script these 4 lines start with `#` (switched off), so they don't reinstall every time. That's on purpose.

3. Open the script and run it

1. *File* → *Open File...* → Documents → ETH → HS26 → Experimental design → Ex3 → `Ex3_Alexander_Konvalina.R`.
2. Press **Cmd + Shift + Enter** (that's *Source with Echo*: R runs the whole script and prints all results in the console). It takes about 10–20 seconds, because `confint(mix2)` is slow.
3. Scroll through the console: there must be **no line starting with** `Error in`. Two plots for Q1 and the BLUE/BLUP plot appear in the Plots pane (flip with the arrows).

If you want to watch it step by step instead, click into the first line and keep pressing **Cmd + Enter**.

TRAP **Red text that is NOT a problem**

The console shows a few scary-looking lines. All of these are normal:

You see…	What it means
`Loading required package: Matrix`, `Attaching package: 'lmerTest'` … `masked`, `Welcome to emmeans`	start-up info of the packages
`Error: MainPlot` and `Error: Within` (in `summary(lm.good)`)	**not errors!** These are the headings of the two parts of the ANOVA table, because the model has `Error(MainPlot)` in it
`1 observation deleted due to missingness`	one yield value is missing (`NA`, LH38 under low N, replicate 1). Mentioned in the data file
`NOTE: Results may be misleading due to involvement in interactions`	expected, the exercise sheet explains it (section 6.4.3)
`boundary (singular) fit: see help('isSingular')`	expected: the genotype-by-N variance is estimated as exactly 0 (that's your Q4/Q5 answer)
`Computing profile confidence intervals ...`	`confint(mix2)` is working, just wait

4. The answers (already in the script)

Nothing to do here. This section shows what each answer block in the script says, so you know what you're handing in. The numbers are from my test run; your output should show the same.

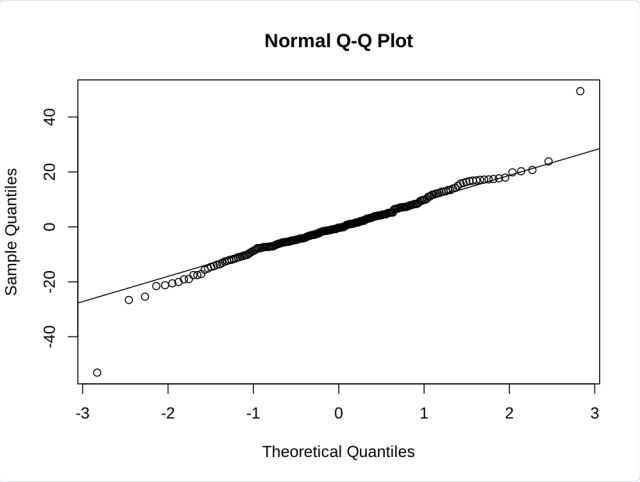
Q1 · 2 points

Extract the residuals, make a normal Q-Q plot and a residuals-vs-fitted plot. Is there a strong deviation from normality or homoscedasticity?

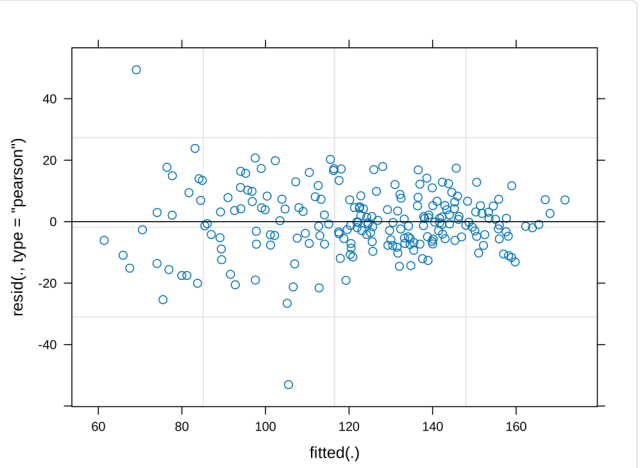
IN THE SCRIPT:

```
# Q1 (2p)
res <- residuals(mix1)
qqnorm(res)
qqline(res)
plot(mix1)
# in the Q-Q plot most points are on the line, only at the two ends some points go
# away from it. especially 2 outliers, one very low and one very high residual
# (both B97 x UH007 under low N). so the residuals are roughly normal distributed.
# in the residuals vs fitted plot the points are spread around 0 without a clear
# pattern. the spread is a bit bigger for low fitted values (low N) than for high
# ones, but its not a strong funnel.
# so i dont see a strong deviation from normality and homoscedasticity, apart from
# the 2 outliers.
```

`qqnorm` and `qqline` draw the Q-Q plot, and `plot(mix1)` draws the residuals against the fitted values (as the question says). The two outliers are the far-away points at the bottom left and top right of the Q-Q plot: B97 under low N in replicate 2 (52.47 dt/ha, very low) and in replicate 3 (118.51, very high).



`qqnorm(res)` + `qqline(res)`



`plot(mix1)`

The two Q1 plots, as they appear in your report.

Q2 · 2 points

Why are A310, NLevelH and Replicate1 missing in `Fix`, and what would their effect be? Then calculate the yield of A310 under low N and EC169 under high N.

IN THE SCRIPT:

```
# Q2 (2p)
# A310 x UH007, NLevelH and Replicate1 are the first levels of their factors
# (alphabetical order). R sets the effect of the first level to 0 and estimates all
# other levels relative to it, so they dont show up in Fix. if they were listed,
# their effect would be 0.
# (so the intercept 134.95 is the yield of A310 under high N in replicate 1)
# A310 x UH007 under low N: intercept + NLevelL + replicate
# (genotype and interaction effect are 0, because A310 is the first level)
134.95 + (-25.16) + (-6.68)
# EC169 x UH007 under high N: intercept + genotype effect + replicate
# (NLevel and interaction effect are 0, because H is the first level)
134.95 + 19.49 + (-6.68)
# so A310 under low N = 103.11 dt/ha and EC169 under high N = 147.76 dt/ha.
# these are the same values as in MeansPT further down.
```

The numbers come from `Fix`: intercept 134.95, `NLevelL` −25.16, `GenotypeEC169 x UH007` 19.49. The −6.68 for the replicates is given in the question. Check: `MeansPT` shows exactly 103.11 (A310, L) and 147.76 (EC169, H).

Q3 · 1 point

What is the (meaningfully rounded) mean of the 6 main-plot effects, and why was this expected?

IN THE SCRIPT:

```
mean(unlist(Ran$MainPlot))

# Q3 (1p)
# R gives a really tiny number (e-13), so rounded the mean of the main-plot effects
# is 0. this was expected because random effects are assumed to follow a normal
# distribution with mean 0 (E(u) = 0). so the main-plot effects are only deviations
# around 0 and they cancel each other out (e.g. 3.16 and -3.16).
```

My R printed `-3.187082e-13`. Yours may show a slightly different tiny number (for example `1.2e-14`). The answer stays the same: rounded, it is 0.

Q4 · 1 point

What is the variance due to genotype, genotype-by-fertilizer interaction, main plot and residual?

IN THE SCRIPT:

```
Var

# Q4 (1p)
# genotype: 54.5
# genotype-by-fertilizer interaction: 0
# main-plot: 208.1
# residual error: 167.5
# (variances in (dt/ha)^2)
```

The numbers come from `Var`: `0.0000 54.4766 208.1046 167.4519`. The exercise sheet shows 54.47751 / 208.11348 / 167.45150, because a slightly older lme4 version was used. Rounded to one decimal it is the same, which is why the answer uses one decimal.

Q5 · 2 points

Fill in the table with σ and σ² (estimates and confidence intervals). Which variances are significant? Same result as when genotype was fixed?

IN THE SCRIPT:

```
CI <- confint(mix2)
CI
# CIs squared, to get the CIs of the variances (the first 4 rows are the sigmas)
CI[1:4, ]^2

# Q5 (2p)
#                     sigma                      sigma^2
#                     estimate  conf. int.      estimate  conf. int.       signif.?
# Genotype             7.38     4.94 - 10.40     54.5     24.4 - 108.2     yes
# Genotype-by-NLevel   0        0    - 4.63       0        0   - 21.5      no
# Main-Plot           14.43     4.81 - 16.92    208.1     23.1 - 286.2     yes
# Residual            12.94    11.69 - 14.43    167.5    136.7 - 208.3     yes
#
# a variance is significant if its CI doesnt include 0. so genotype and main-plot
# (and the residual) are significant, genotype-by-NLevel is not, its CI starts at 0
# (and the estimate is 0).
# compared to anova(mix1) where genotype and genotype:NLevel were fix: genotype was
# significant there too (p = 1.2e-05) and genotype:NLevel was not significant there
# too (p = 0.86). so yes, the result is the same for both.
```

Where the numbers come from: **σ estimates** from `VarCorr(mix2)` (7.3808, 0.0000, 14.4258, 12.9403), **σ CIs** from `CI` (the exercise says `.sig01` = Genotype:NLevel, `.sig02` = Genotype, `.sig03` = MainPlot, `.sigma` = Residual), **σ² estimates** from `Var` , **σ² CIs** from `CI[1:4, ]^2` . If a last digit in your output differs (for example 21.4 instead of 21.5), change it in the table.

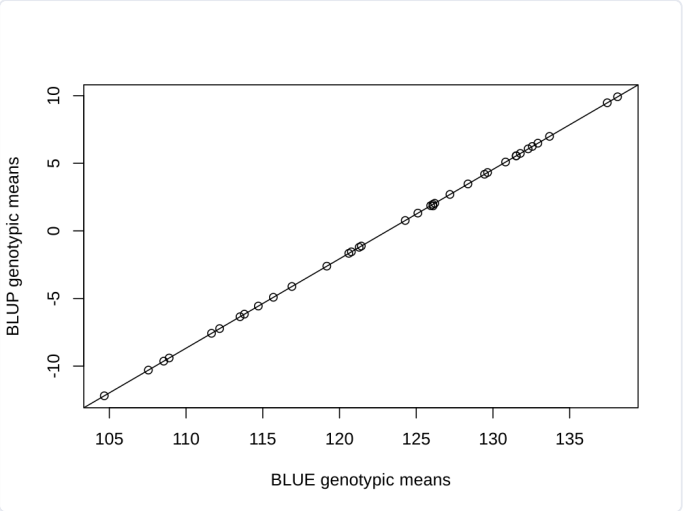
Q6 · 2 points

How good is the fit of the BLUP-on-BLUE regression, what is the slope, and what are the ranges (max – min) of BLUEs and BLUPs? Interpret R^2 and the slope.

IN THE SCRIPT:

```
# Q6 (2p)
min(GFix)
max(GFix)
max(GFix) - min(GFix)
min(GRan)
max(GRan)
max(GRan) - min(GRan)
(max(GRan) - min(GRan)) / (max(GFix) - min(GFix))
# the fit is basically perfect, R-squared is 1 and all points are on the line.
# the slope is 0.661.
# BLUEs go from 104.67 to 138.12 -> difference 33.45
# BLUPs go from -12.20 to 9.92 -> difference 22.12
# R2 = 1 means BLUPs and BLUEs are perfectly linearly related, so the ranking of the
# genotypes is the same for both. the BLUPs are just the BLUEs shrunken towards
# the mean.
# the slope 0.661 is the shrinking factor (it is the heritability,
# h2 = 54.47 / 82.38 = 0.661). the range of the BLUPs is 0.661 times the range of
# the BLUEs (22.12 / 33.45 = 0.661), so the BLUPs are closer together than the BLUEs.
```

R^2 and the slope come from `summary(fit)` : `Multiple R-squared: 1` and `GFix 6.611e-01` . The ranges come from the `min` / `max` lines right above the answer. The last line, 22.12 / 33.45, shows that the ratio of the ranges is exactly the slope. The heritability 54.47 / 82.38 is taken from the exercise sheet.



BLUPs against BLUEs, as in your report. All points lie on the line: R^2 = 1, slope 0.661.

5. Make the PDF and hand it in

1. **Save** the script (Cmd + S).
2. *File → Compile Report…* (or **Cmd + Shift + K**). Choose **HTML** and click **Compile**. If RStudio asks to install or update packages, click **Yes**. After a few seconds a preview opens, and `Ex3_Alexander_Konvalina.html` appears in the `Ex3` folder.
3. In the preview, click **Open in Browser** (Safari).

4. Press **Cmd + P**. Untick **"Print headers and footers"** (if you don't see it, click *Show Details*). Then click the **PDF** menu at the bottom left → **Save as PDF** and save it as `Ex3_Alexander_Konvalina.pdf`.
5. **Check the PDF:** title "Exercise 3 - Mixed Models" and your name on page 1, about 22 pages, the 3 plots are there, and nowhere a line starting with `## Error in`.
6. **Upload** the PDF to the Exercise 3 repository on Moodle, before 05.11.2026, 23:59.

> **TIP** **Why is the report so long?**
>
> `summary(mix1)`, `Fix` and `ranef(mix2)` print long lists (36 genotypes, 72 genotype × N combinations). That's what the exercise code does, so it's fine. The answers are the `#` comments right below each `# Q... (…p)` line.

6. If something goes wrong

You see…	Do this
`Error in setwd(...) : cannot change working directory`	The folder is not exactly `Documents/ETH/HS26/Experimental design/Ex3`. Check the name (capital E in `Ex3`), or change the path in the `setwd` line to where your folder is.
`cannot open file 'DelleyDef.txt': No such file or directory`	The data file is not in `Ex3`, or it has a different name. Rename it to exactly `DelleyDef.txt`.
`there is no package called 'lme4'` (or `lmerTest`, `emmeans`, `pbkrtest`)	Install it, see section 2.
`could not find function "lmer"`	The `library(...)` lines did not run. Press Cmd + Shift + Enter again (it runs everything from the top).
`anova(mix1)` shows no `Pr(>F)` column	`lmerTest` was loaded after the model was made. *Session* → *Restart R*, then Cmd + Shift + Enter.
Files show a small cloud icon in Finder	Your Documents folder is in iCloud. Right-click `Ex3` → *Download Now*.
You don't have `DelleyDef.txt`	Replace the `read.table` line with the one from the exercise sheet, which reads the file from the internet: `Data <- read.table("https://edu.usys.ethz.ch/statdat/DelleyDef.txt", sep = "\t", skip = 5, header = TRUE)`.
Anything weird	*Session* → *Restart R*, then Cmd + Shift + Enter.

Tested by running the finished script (whole script, and with Compile Report) inside a copy of the folder Documents/ETH/HS26/Experimental design/Ex3, with R 4.6.1 and the newest packages (lme4 2.1.0, lmerTest 3.2.1, emmeans 2.0.4, pbkrtest 0.5.5, Matrix 1.7.6). All numbers above come from that run. The internet link from the exercise sheet could not be tested.